PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

DigiTrack: An Instructor Scheduling and Performance Monitoring System for Driving Schools

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1: INTRODUCTION

1.1 BACKGROUND OF THE STUDY

In the modern era of rapid urbanization, the demand for structured and professional driver education has become a cornerstone of public safety. In the Philippines, driving is an essential life skill, with cars and motorcycles serving as primary modes of transportation. According to Statista Research Development, the number of registered private vehicles in the country is approximately 1.27 million. As the number of Filipinos acquiring vehicles for personal and business use increases, the demand for skilled drivers rises, necessitating a higher standard of professional driving education.

Despite this growth, many institutions continue to rely on traditional, paper-based operations. Hortaleza et al. (2024) and Khanvailkar et al. (2022) observe that driving schools still depend on manual methods for data management, leading to communication breakdowns and clerical errors. Relying on phone calls or physical visits for registration is time-consuming and prone to scheduling conflicts. Furthermore, Sudarmilah & Wahendra (2021) note that manual record-keeping is susceptible to data loss, whereas web-based systems can secure student data and provide transparent information regarding training packages and pricing.

The transition to digital operations is a global necessity. Mithrasena (2020) argues that manual processes result in a "waste of human power and money." However, he also notes that many current systems are limited to basic registration. There is a clear need for "feature-rich" solutions that include advanced monitoring. Kulshrestha et al. support this by highlighting that centralized monitoring systems enhance safety through real-time oversight.

For Motosafe Driving School, these challenges are a daily reality. The lack of a centralized Management Information System (MIS) leads to uneven instructor workloads and a lack of objective performance data. The proposed system, DigiTrack, seeks to bridge these gaps. By synthesizing the administrative efficiency of automated scheduling with a robust Performance Monitoring Layer, DigiTrack will modernize Motosafe’s operations, ensuring accountability and improving the quality of driver education.

1.2 STATEMENT OF THE PROBLEM

GENERAL PROBLEM STATEMENT:

Motosafe Driving School faces operational inefficiencies due to manual scheduling and a lack of centralized instructor monitoring, resulting in uneven workloads and difficulty in maintaining consistent training quality.

SPECIFIC PROBLEMS:

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1.3 OBJECTIVE OF THE STUDY

1.3.1 GENERAL OBJECTIVE

To develop and implement **DigiTrack**, a web-based scheduling and performance monitoring system designed to streamline instructor management and optimize workload distribution for Motosafe Driving School.

1.3.2 SPECIFIC OBJECTIVE

- To design an automated scheduling module that prevents conflicting appointments and overlaps.
- To develop a workload distribution algorithm that assigns students to instructors based on real-time availability and utilization rates.
- To create a student-facing feedback mechanism to track and store instructor performance metrics.
- To implement a payment management module for tracking student transactions and financial records.
- To provide an administrative dashboard for real-time visualization of instructor utilization and student outcomes.

SCOPE:

The system is a web-based application accessible to three user roles: Admin, Instructor, and Student. It features automated booking, workload balancing, a feedback mechanism, and a payment tracking module. The system will generate real-time reports on instructor utilization and student progress.

DELIMITATION:

The system will not calculate government taxes or official salaries. It does not integrate with external government databases (e.g., LTO) or use GPS/vehicle sensors; it relies on manual web-based data entry. Student ratings are for Motosafe's internal quality control and are not official legal certifications.

1.4 SIGNIFICANCE OF THE STUDY

The development of the DigiTrack Web-Based Scheduling and Performance Monitoring System will provide benefits to various groups involved in the driving school operations:

- Administrators: This system will enable administrators to efficiently manage instructor schedules, monitor instructor performance, and generate real-time reports. It will streamline administrative processes, reduce scheduling conflicts, and improve overall operational efficiency.

- Instructors: Instructors will benefit from a balanced workload facilitated by automated scheduling. The performance monitoring feature will help instructors receive feedback, identify areas for improvement, and enhance their teaching effectiveness.

- Students: Students will experience more organized and timely training sessions, as well as the opportunity to provide feedback on their instructors, leading to improved service quality and satisfaction.

- Driving School Operations: Implementing this system will modernize the management process, promote accountability, and ensure compliance with training standards, ultimately improving the quality of driver education.

- Researchers and Future Developers: This project can serve as a reference for future research on automated scheduling and performance monitoring systems in educational or training institutions.

- Academic Community: This study contributes to the fields of Information Technology and Management by demonstrating how web-based systems can optimize instructor management and improve service delivery in driving schools.

1.5 DEFINITION OF TERMS

- DigiTrack: A web-based scheduling and performance monitoring system designed for driving schools to streamline instructor management, track student outcomes, and improve operational efficiency.

- Manual Scheduling: A traditional approach where appointments and instructor assignments are arranged manually, often using paper-based methods or basic spreadsheets, which can lead to scheduling conflicts and inefficiencies.

- Workload Distribution: The process of allocating teaching hours and responsibilities equally among instructors to ensure balanced workloads and optimal utilization.

- Performance Monitoring: The continuous assessment of instructors' teaching effectiveness and adherence to training standards through feedback, evaluations, and data analysis.

- Conflict-Free Scheduling: An automated scheduling method that prevents overlapping bookings and ensures that instructors are assigned to sessions without conflicts.

- Instructor Utilization: The extent to which instructors' available hours or sessions are effectively scheduled and used within a given period.

- Student Feedback: Ratings and comments provided by students regarding their training sessions and instructors are used to evaluate instructor performance and improve service quality.

- Automation: The use of software to perform tasks automatically, reducing manual effort and minimizing human errors in scheduling, monitoring, and reporting processes.

- Performance Metrics: Quantitative measures used to evaluate instructor effectiveness, student satisfaction, and overall system performance.

- Administrative Dashboard: A centralized interface for administrators to manage schedules, monitor instructor performance, view reports, and oversee system operations efficiently.

- Feedback System: A feature that allows students to rate and comment on their training experiences, providing valuable insights for quality improvement.

- Web-Based System: A software application stored on a remote server and delivered over the Internet through a browser interface.

CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

2.1 FOREIGN STUDIES

According to the study TrainMe: A Web-Based Management System for Driving Schools by A.H.R.P Mithrasena (2020), manual processes result in a "waste of human power and money" in the modern information technology era. The author argues that a "feature-rich" digital solution must include not only registration but also vehicle tracking and online payment facilities to deliver maximum value to the customer. He supports the architectural design of DigiTrack, which adopts the three-user level (Admin, Instructor, and Student) to manage limited resources efficiently while addressing the qualitative gaps identified in previous manual systems.

Furthermore, the study Centralized Driving Schools Monitoring System by Arjun Kulshrestha et al. highlights the importance of "driving behavior" analytics, such as speed, acceleration, and braking, to enhance the safety and security of both drivers and passengers. The system provides the Regional Transport Office with a submission portal report for daily feedback, ensuring quality education through real-time oversight. This justifies the inclusion of a Performance Monitoring Layer in the current project, moving beyond administrative tasks toward data-driven quality assurance.

2.2 LOCAL STUDIES

According to the study DRIVRHUB: A Multi-Platform Driving School Management System by Royce Hortaleza et al. (2024), conducted in Pangasinan, the success of Filipino driving schools depends on "clarity and responsiveness" during the registration and booking processes. The researchers found that manual delays in session scheduling often led to a loss of trust from student drivers. To address these challenges, they proposed a system that manages Registration, Payment, and Report Generation, ensuring that automation effectively meets the evolving needs of local driving schools. This directly supports DigiTrack, as it validates that a centralized system is a core requirement for many modern driving schools operating in the country.

2.3 RELATED LITERATURE

2.3.1 Information System and Management

According to Management Information Systems: Managing the Digital Firm by Laudon and Laudon (2020), a Management Information System (MIS) is a set of interrelated components that collect, process, store, and distribute information to support decision-making in an organization. For Motosafe Driving School, DigiTrack functions as an MIS by transforming manual student logs into digital data. This allows administrators to monitor Instructor Utilization and Workload Balancing, ensuring that resources are distributed efficiently.

2.3.2 Software Development Methodology: Agile Scrum

The development of DigiTrack follows the Agile Scrum Methodology. Agile is effective for system development because it supports iterative development and continuous improvement.

2.3.3 Web-Based Technology and Architecture

DigiTrack utilizes a Client-Server Architecture, which is the standard for web applications

Frontend Technologies: These are tools used to develop the user interface. This includes HTML, CSS, Bootstrap, and JavaScript.

Backend and Databases: The system uses MySQL as a database and PHP as the backend.

2.4 SYNTHESIS OF THE REVIEWED LITERATURE

The reviewed foreign and local studies show that driving school operations can be significantly improved through the use of digital technologies. Research highlights that manual scheduling processes are often inefficient and prone to human error, leading to conflicts and uneven instructor workload.

Furthermore, the literature highlights the importance of automated systems in improving management efficiency, monitoring instructor performance, and providing better service to students. These finding supports the development of DigiTrack as a web-based solution that aims to streamline scheduling, enhance monitoring, and improve overall operational management in driving schools.

Research Gaps:

While existing studies focus on general “school finders” or expensive technologies like GPS tracking, there is a lack of specialized, low-cost management tools that focus on internal instructor workload balancing and performance monitoring within a single institution.

DigiTrack addresses these gaps by offering a web-based system that automates instructor scheduling to prevent conflicts and evenly distribute workloads. It also includes a feedback system, performance monitoring, and dashboard for real-time monitoring, enabling efficient data-driven management without the need for expensive hardware.

CHAPTER 3: METHODOLOGY

3.1 RESEARCH DESIGN

This study uses Developmental Research because the primary goal is to create, test, and evaluate DigiTrack to solve the manual scheduling of Motosafe Driving School. It also incorporates descriptive research to analyze the current manual workflow through interviews and observations.

3.2 SYSTEM DEVELOPMENT METHODOLOGY

This project follows the Agile Scrum Model, which will allow developers to work in "sprints". It includes the following phase:

- Requirement Analysis Phase
- System Design
- Development
- Testing
- Deployment

3.3 SYSTEM ARCHITECTURE

DigiTrack follows a Client-Server Architecture.

- Client Side: Web interface where Admins, Instructors, and Students interact with the system via browser.
- Server Side: Processes the request and ensures no conflicts.

3.4 TOOLS AND TECHNOLOGY USED

- Frontend: HTML, CSS. JavaScript, Bootstrap (for responsive user interface)
- Backend: PHP
- Database: MySQL
- Development Tools: Visual Studio Code and XAMPP/Apache

3.5 DATA GATHERING TECHNIQUES

- Interviews
- Observations

3.6 TESTING PROCEDURES

- Unit Testing: Testing individual features to ensure they work.
- System Testing: Testing how system module interacts with the database.

3.7 EVALUATION CRITERIA

- Functionality
- Usability
- Reliability
- Security

REFERENCES:

Foreign Studies

Kulshrestha, A., et al. (2020). *Centralized Driving Schools Monitoring System*. International Journal of Engineering Research & Technology (IJERT).

Mithrasena, A. H. R. P. (2020). *TrainMe: A Web-Based Management System for Driving Schools*. Sri Lanka Institute of Information Technology.

Local Studies

Hortaleza, R., et al. (2024). *DRIVEHUB: A Multi-Platform Driving School Management System with Driving School Finder*. [Unpublished Capstone Project/Journal], Pangasinan, Philippines.

Books and Literature

Laudon, K. C., & Laudon, J. P. (2020). *Management Information Systems: Managing the Digital Firm* (16th ed.). Pearson Education.